DTIE | DEPARTMENT OF MULTI DISCIPLINARY INNOVATION & ENTREPRENEURSHIP

REAL LIFE SOLUTIONS FOR WASTE MANAGEMENT

Design Thinking for Innovation & Entrepreneurship (DTIE) — Section: 1 Project Batch-4

Project Overview & Objectives

This project addresses one of the most pressing municipal and environmental challenges of modern communities: systematic, affordable, and practical waste management. By combining design thinking principles with community-level execution, we aim to transform waste disposal into a sustainable resource cycle.

Team Members

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Core Vision

To create a clean, green, and sustainable environment by reducing waste and encouraging responsible waste disposal across homes, educational institutions, and public communities.

Mission & Problem Statement

Mission Objectives

- Promote systematic waste segregation right at the source.
- Encourage recycling, reuse, and upcycling of materials.
- Convert organic food and garden waste into rich compost.
- Drastically reduce reliance on single-use plastics and disposables.
- Spread grassroots awareness about sustainable waste management practices.
- Develop affordable, low-barrier tools for homes, colleges, and local communities.

The Core Challenge

Waste management is a critical bottleneck in homes, academic campuses, markets, and public squares. Improper disposal triggers environmental pollution, foul odors, blocked drainage channels, and severe public health risks.

Identified Key Pain Points:

- Indiscriminate mixing of wet, dry, and hazardous waste.
- Severe lack of foundational awareness regarding segregation.
- Excessive consumption of single-use plastics.
- Scarce local recycling and composting infrastructure.
- Widespread littering in municipal streets and public spaces.
- Inefficient collection schedules and disposal pathways.

Design Challenge Statement

"How can we reduce waste generation, improve source segregation, and engineer practical solutions that make everyday waste management effortless for individuals and communities?"

Research, Market & Target Audience

Research & Industry Study

Sources: CPCB, MoEFCC, and municipal reports.

- **Current Situation:** Continuous daily waste generation across households, offices, markets, and educational institutions.
- **Processing Hurdles:** Mixed waste complicates recycling, whereas segregated organic streams yield valuable compost.
- **Market Opportunities:** Door-to-door collection, compost retailing, recycling collection points, and eco-alternatives.

Target Users & Needs

- **Primary Users:** Households, apartments, schools, colleges, restaurants, and grocery shops.
- **Core Needs:** Seamless segregation workflows, affordable collection channels, odor control, and clean surroundings.
- **Key Beneficiaries:** Citizens, sanitation workers, local authorities, and the environment.

Competitive Landscape & Opportunity

Existing Solution	Current Limitation	Our Advantage
Municipal Collection	Infrequent schedules, mixed waste intake	Community-level pre-segregation
Scrap Dealers & Recyclers	Limited material categories accepted	Comprehensive multi-stream management
Smart Bins & Apps	High cost, complex maintenance	Affordable, low-maintenance setup

Risk Analysis & Proposed Solution

Risks & Existing Gaps

- Improper separation by untrained users.
- Bad odors resulting from poor organic storage.
- Contamination of otherwise recyclable materials.
- High initial equipment costs or lack of maintenance funds.
- Hazardous risks from batteries and electronic waste.

Mitigation: Color-coded bins, clear instructional labels, authorized recycling partnerships, and routine monitoring.

Our Solution: Smart & Sustainable System

An integrated, community-ready waste management architecture:

- **Green Bin:** Wet / biodegradable organic waste.
- **Blue Bin:** Dry and clean recyclable materials.
- **Red / Special Bin:** Hazardous and electronic waste.
- **Composting Unit:** Turning food scraps into fertilizer.
- **Awareness Drives:** Posters, workshops, and clear signage.

Operational Workflow (Storyboard)

Step 1: Generation
Waste produced at homes & campuses

Step 2: Segregation
Sorting into Green, Blue & Red bins

Step 3: Processing
Composting, recycling & safe disposal

Financial Estimation & Lean Canvas

Pilot Budget (Small College / Community)

Item Description	Cost (INR)
Colour-coded bins (Set of 3)	₹3,000
Composting unit setup	₹4,000
Posters & awareness materials	₹1,000
Gloves & safety equipment	₹1,000
Transportation & setup	₹3,000
Miscellaneous expenses	₹1,000
Total Estimated Capital	**₹13,000**

Lean Canvas Highlights

Customer Segments: Households, apartments, colleges, and local businesses.

Value Proposition: Affordable, simple, eco-friendly community waste handling.

Revenue Streams: Compost sales, recyclable materials, and service fees.

Key Metrics: Waste segregated volume, recycling rate, and active participation.

Funding Sources: College grants, community sponsorships, and municipal initiatives.

Project Implementation Roadmap

Week 1: Research
Study local waste problems & conduct audits.

Week 2: Planning
Select bin types & prepare budget.

Week 3-4: Pilot & Test
Deploy bins, composting & evaluate feedback.

Conclusion & Guidelines

> *“Real-life solutions for waste management begin with simple, responsible actions—separating waste, reusing materials, recycling, and composting organic waste. Our project builds a complete community ecosystem rather than just providing a single bin. Reduce waste, reuse resources, and recycle for a cleaner, greener future.”*

Template Guidelines & Faculty Notes

- **Standard Format:** Prepared in accordance with the DTIE standard template and structural presentation outlines.
- **Creativity & Flexibility:** Subheadings and visual modules have been adapted for clarity, visual engagement, and professional presentation standards.
- **Faculty Support:** For further inquiries regarding implementation, funding, or evaluation, please contact your respective DTIE faculty advisor.

Design Thinking for Innovation & Entrepreneurship (DTIE)

Academic Year 2026-27 | End Semester Project Presentation

Source Code

Application Frontend & Styles Implementation

The following snippet contains the structural markup and Tailwind CSS declarations for the web application interface:

```
<!DOCTYPE html>
<!-- saved from url=(0094)https://app-wyirhme6jcjtlde4qgm2bkqqsl35ryyvbqc3d72cbqljwst4hqm3.makeproxy-c.figma.site/#start -->
<html lang="en"><head><meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <script type="module">import { injectIntoGlobalHook } from "/@react-refresh";
injectIntoGlobalHook(window);
window.$RefreshReg$ = () () => {};
window.$RefreshSig$ = () => (type) => type;</script>

    <script type="module" src="./client"></script>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Figma Make App</title>
    <link rel="icon" href="https://app-wyirhme6jcjtlde4qgm2bkqqsl35ryyvbqc3d72cbqljwst4hqm3.makeproxy-c.figma.site/favicon.svg">
    <meta property="og:title" content="Figma Make App">
  <style type="text/css" data-vite-dev-id="/workspaces/default/code/src/index.css">
@import url("https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700&family=Space+Grotesk:wght@600;700&display=swap");
@layer properties;
@layer theme, base, components, utilities;
/*! tailwindcss v4.2.2 | MIT License | https://tailwindcss.com */
@layer theme {
  :root, :host {
    --font-sans: "Plus Jakarta Sans", sans-serif;
    --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, "Liberation Mono",
      "Courier New", monospace;
.backdrop-blur-md {
    --tw-backdrop-blur: blur(var(--blur-md));
    -webkit-backdrop-filter: var(--tw-backdrop-blur,) var(--tw-backdrop-brightness,) var(--tw-backdrop-contrast,) var(--tw-backdrop-grayscale,) var(--tw-backdrop-hue-rotate,) var(--tw-backdrop-invert,) var(--tw-backdrop-opacity,) var(--tw-backdrop-saturate,) var(--tw-backdrop-sepia,);
    backdrop-filter: var(--tw-backdrop-blur,) var(--tw-backdrop-brightness,) var(--tw-backdrop-contrast,) var(--tw-backdrop-grayscale,) var(--tw-backdrop-hue-rotate,) var(--tw-backdrop-invert,) var(--tw-backdrop-opacity,) var(--tw-backdrop-saturate,) var(--tw-backdrop-sepia,);
  }
  .backdrop-blur-xl {
    --tw-backdrop-blur: blur(var(--blur-xl));
    -webkit-backdrop-filter: var(--tw-backdrop-blur,) var(--tw-backdrop-brightness,) var(--tw-backdrop-contrast,) var(--tw-backdrop-grayscale,) var(--tw-backdrop-hue-rotate,) var(--tw-backdrop-invert,) var(--tw-backdrop-opacity,) var(--tw-backdrop-saturate,) var(--tw-backdrop-sepia,);
    backdrop-filter: var(--tw-backdrop-blur,) var(--tw-backdrop-brightness,) var(--tw-backdrop-contrast,) var(--tw-backdrop-grayscale,) var(--tw-backdrop-hue-rotate,) var(--tw-backdrop-invert,) var(--tw-backdrop-opacity,) var(--tw-backdrop-saturate,) var(--tw-backdrop-sepia,);
  }
  .transition {
    transition-property: color, background-color, border-color, outline-color, text-decoration-color, fill, stroke, --tw-gradient-from, --tw-gradient-via, --tw-gradient-to, opacity, box-shadow, transform, translate, scale, rotate, filter, -webkit-backdrop-filter, backdrop-filter, display, content-visibility, overlay, pointer-events;
    transition-timing-function: var(--tw-ease, var(--default-transition-timing-function));
    transition-duration: var(--tw-duration, var(--default-transition-duration));
  }
  .transition-transform {
    transition-property: transform, translate, scale, rotate;
    transition-timing-function: var(--tw-ease, var(--default-transition-timing-function));
    transition-duration: var(--tw-duration, var(--default-transition-duration));
</style>
```